Installation Guide

From zero to your first notebook, step by step

Before you start

This guide takes about **30 minutes**. You don't need any prior knowledge. Follow the steps in order and don't skip any.

You need:

- Your **laptop** (Windows or macOS) with internet access
- The **password** of your laptop, because installing software sometimes asks for it

At the end, you will have everything you need for the tutorials.

The big picture: five things you will use

What	In simple words	Why we need it
Python	A programming language	We write our programs in Python
VS Code	An app to write code, like Word for text	This is where you write and run your code
Terminal	A window where you type commands instead of clicking	We use it to install things
uv	A small helper tool	It installs Python and everything Python needs
Notebook	A document that mixes text and code you can run	All tutorials are notebooks

How they fit together: you open **VS Code**, use its **terminal** to let **uv** install **Python**, and then work in **notebooks** inside VS Code.

Step 1: Install VS Code

Where: your web browser

Why: VS Code is the app you will work in for the whole course. Apps like this are called IDEs (Integrated Development Environments): one place to write code, run it and find mistakes.

Do: Go to code.visualstudio.com and download VS Code for your system.

macOS

1. Open the downloaded file. A file called **Visual Studio Code** appears.
2. Drag it into your **Applications** folder.
3. Open it from Applications. If your Mac asks whether you want to open an app from the internet, click **Open**.

Windows

1. Open the downloaded installer.
2. Accept the agreement and keep the default settings. Make sure **"Add to PATH"** is ticked.
3. Click **Install**, then **Finish**.

Check: VS Code opens and shows a welcome page.

Step 2: Add the Python and Jupyter extensions

Where: VS Code

Why: Out of the box, VS Code doesn't know Python or notebooks. Extensions are small add-ons that teach it.

Do:

1. Click the **Extensions** icon on the left side (four small squares).
2. Search for **Python** and click **Install** on the one by Microsoft.
3. Search for **Jupyter** and click **Install** on the one by Microsoft.

Check: Both extensions show "Installed" or a gear icon instead of an Install button.

Step 3: Create your course folder

Where: Finder (macOS) or File Explorer (Windows)

Why: Everything for this course lives in one folder. That keeps things tidy, and Python works best when each project has its own folder.

macOS

Open **Finder**, go to your **home folder** (the one with the house icon) and create a new folder called `programming-course`.

Windows

Open **File Explorer**, go to `C:\Users\YourName` and create a new folder called `programming-course`.

Please **don't** put it in a OneDrive folder. Syncing can break the setup.

Do in VS Code: Click **File → Open Folder...**, select `programming-course` and click **Open**. If VS Code asks whether you trust the authors of the files, click **Yes, I trust the authors**.

Check: On the left side of VS Code, you see the name `PROGRAMMING-COURSE` at the top.

Step 4: Open the terminal

Where: VS Code

Why: The terminal is where you type commands. You type a command, press **Enter**, and the computer answers with text. We only need a few commands, and you can copy them from this page.

Do: In the menu bar, click **Terminal → New Terminal**. A panel opens at the bottom of VS Code.

macOS

The terminal shows a line ending with `%`. That's where you type.

Windows

The terminal shows a line starting with `PS` and ending with `>`. `PS` means **PowerShell**, which is exactly what we need. If you don't see `PS`, click the small arrow next to the **+** in the terminal panel and choose **PowerShell**.

Good to know:

- Copy a command from this page, paste it into the terminal and press **Enter**.
- Paste with **Cmd + V** (macOS) or **Ctrl + V** or a right-click (Windows).
- Some commands take a few seconds. Wait until the line with `%` or `>` is back before typing the next one.

Check: You see a terminal at the bottom of VS Code, and its line mentions `programming-course`.

Step 5: Install uv

Where: the terminal in VS Code

Why: uv is a helper that installs Python for you and keeps each project clean. You only install it once.

Do: Copy the command for your system and press **Enter**.

macOS

```
curl -LsSf https://astral.sh/uv/install.sh | sh
```

Windows

```
powershell -ExecutionPolicy ByPass -c "irm https://astral.sh/uv/install.ps1 | iex"
```

Now **close VS Code completely** and open it again. This is important: only then does the terminal know the new tool. Open a new terminal again with **Terminal → New Terminal**.

Check: Type this and press **Enter**:

```
uv --version
```

You should see something like `uv 0.11.26`. The exact number doesn't matter.

Step 6: Install Python

Where: the terminal in VS Code

Why: This installs the Python language itself. Everyone in the course uses the same version, so the code works the same for all.

Do:

```
uv python install 3.13
```

Check: The terminal says that Python 3.13 was installed, or that it is already installed. Both are fine.

Step 7: Turn your folder into a Python project

Where: the terminal in VS Code, inside `programming-course`

Why: This tells uv: "this folder is a Python project". uv creates a few files it needs to keep track of things.

Do: Run these two commands, one after the other:

```
uv init
```

```
uv python pin 3.13
```

New files appear on the left in VS Code. You don't need to touch them:

- `main.py`: a tiny example program
- `pyproject.toml`: the list of what your project needs
- `.python-version`: which Python version the project uses

Check: Run the example program:

```
uv run python main.py
```

You should see `Hello from programming-course!`. Congratulations, you just ran your first Python program!

Step 8: Add notebook support

Where: the terminal in VS Code

Why: Notebooks need two extra helpers. The first one lets notebooks use the Python of your project, the second one adds the notebook tools.

Do:

```
uv add --dev ipykernel
```

```
uv add jupyter
```

Check: The terminal lists a lot of installed packages and ends without an error. A new folder `.venv` appears on the left. That's your project's own Python environment.

Step 9: Your first notebook

Where: VS Code

Why: This checks that everything works together, before you start with the tutorials.

Do:

1. Press **Cmd + Shift + P** (macOS) or **Ctrl + Shift + P** (Windows). A search bar opens at the top.
2. Type `Create: New Jupyter Notebook` and press **Enter**.
3. Click **Select Kernel** in the top right corner. The kernel is the Python that runs your notebook.

4. Choose **Python Environments**, then the one that says **.venv** and **Python 3.13**.
5. Click into the first cell, type `print("Hello")` and press **Shift + Enter**.

Check: Below the cell, you see `Hello`. Everything works!

If VS Code asks to install extensions at any point, click **Install**.

Step 10: Open a tutorial

Where: your web browser, then VS Code

Do:

1. Open a tutorial on the course website.
2. On the right side under **Other Formats**, click **Jupyter** to download the notebook.
3. Move the downloaded file into your `programming-course` folder.
4. Open it in VS Code by clicking it on the left.
5. If VS Code asks for a kernel, choose the same **.venv** with **Python 3.13** as in Step 9.

You're ready for the tutorials!

Checklist

- ☐ VS Code is installed and opens
- ☐ The Python and Jupyter extensions are installed
- ☐ You have a folder `programming-course` and opened it in VS Code
- ☐ `uv --version` shows a version number
- ☐ `uv run python main.py` shows `Hello from programming-course!`
- ☐ A notebook cell with `print("Hello")` shows `Hello`

Every next session

You don't need to install anything again. Just:

1. Open VS Code.
2. Click **File → Open Recent → programming-course**.
3. Open the notebook you want to work on.

Something doesn’t work?

Problem	What to do
`uv` is “not found” or “not recognized”	Close VS Code completely and open it again. On Windows, check that the terminal says `PS`.
Windows says scripts are disabled	Run `Set-ExecutionPolicy -ExecutionPolicy Bypass -Scope Process`, then repeat Step 5.
`.venv` doesn’t appear under Select Kernel	Press **Cmd/Ctrl + Shift + P**, type `Developer: Reload Window`, press **Enter**, try again.
The notebook shows a different Python than 3.13	Click on the kernel name in the top right and choose the **.venv** one.
`uv self update` fails on a Mac	If you installed uv with Homebrew, use `brew upgrade uv` instead.
Anything else	Ask us! Guessing usually takes longer than asking.

Later in the course: adding packages

Sometimes a tutorial needs an extra package, for example `pandas`. Then run this in the terminal of your course folder:

```
uv add pandas
```